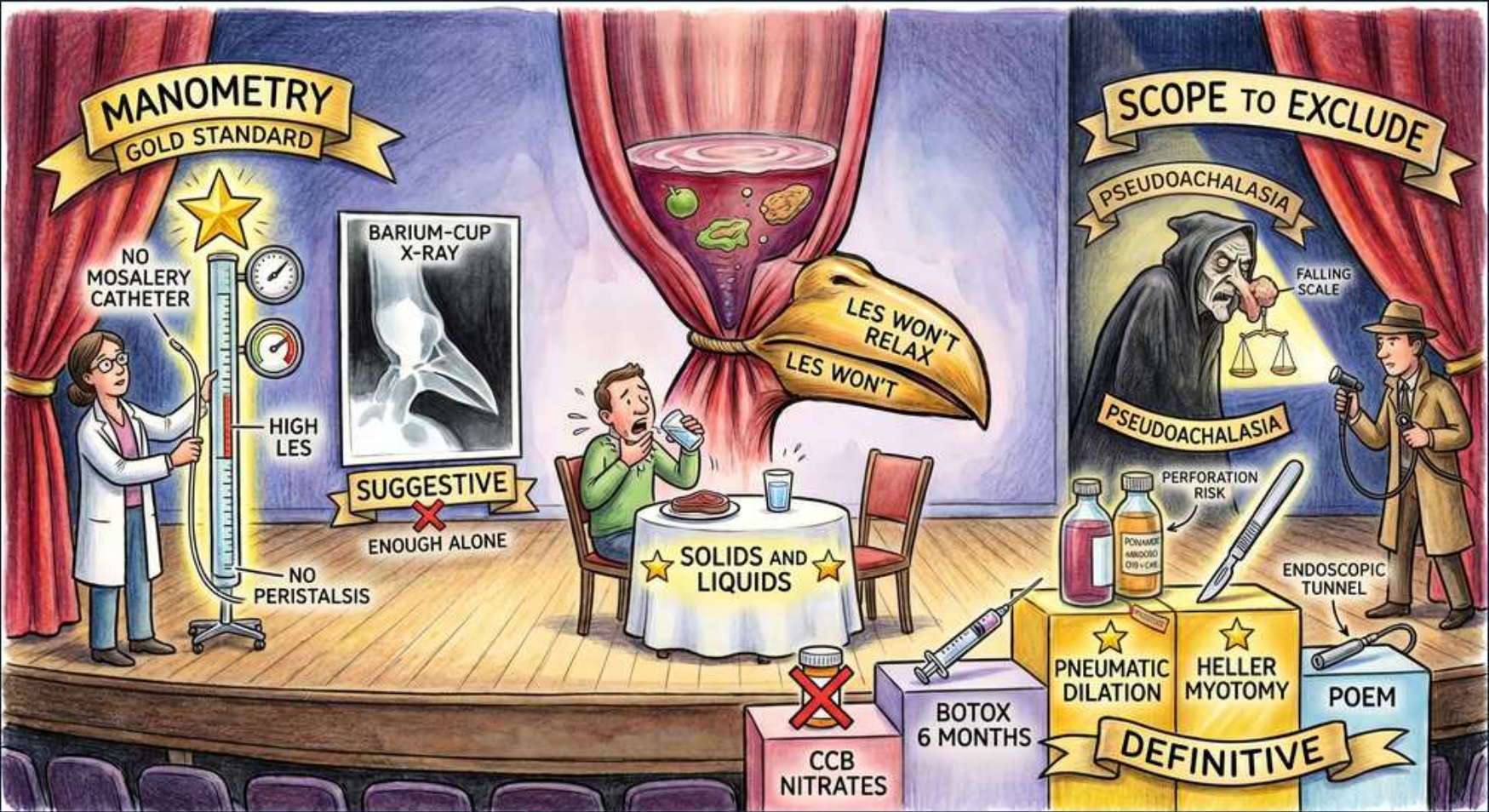


Esophageal — Achalasia



1. Manometry = gold standard

WHAT YOU SEE

Gold-star banner 'MANOMETRY / GOLD STANDARD' over the doctor with a manometry column.

WHAT IT ENCODES

High-resolution esophageal MANOMETRY is the diagnostic gold standard for achalasia: shows elevated LES resting pressure, failure of the LES to relax with swallowing, and absent peristalsis (aperistalsis) in the body. It defines the diagnosis when imaging is suggestive.

2. High LES pressure

WHAT YOU SEE

Pressure gauge marked 'HIGH LES' on the manometry stand.

WHAT IT ENCODES

Manometric hallmark: elevated LES resting pressure with incomplete relaxation (integrated relaxation pressure raised). The LES stays tonically tight, obstructing outflow at the GE junction.

3. No peristalsis

WHAT YOU SEE

Label 'NO PERISTALSIS' at the base of the manometry column.

WHAT IT ENCODES

Loss of ganglion cells in the myenteric (Auerbach) plexus causes aperistalsis of the esophageal body plus the non-relaxing LES. This degeneration of inhibitory (nitric-oxide) neurons is the core mechanism.

4. Dysphagia to solids AND liquids

WHAT YOU SEE

Gold-star banner 'SOLIDS AND LIQUIDS' at the dining table with a struggling diner.

WHAT IT ENCODES

Achalasia causes dysphagia to BOTH solids and liquids from the start = a MOTILITY disorder. Contrast with mechanical obstruction (stricture/cancer), which gives progressive solids-then-liquids dysphagia. Also regurgitation of undigested food, weight loss.

5. LES won't relax

WHAT YOU SEE

Funnel-shaped curtain narrowing at the bottom, banners 'LES WON'T RELAX'.

WHAT IT ENCODES

The non-relaxing LES creates the functional obstruction; food piles up above it and the esophagus dilates over time, giving the dilated body tapering to a tight GE junction.

6. Barium bird's-beak (suggestive only)

WHAT YOU SEE

X-ray film 'BARIUM-CUP X-RAY' with a 'SUGGESTIVE / ENOUGH ALONE (X)' label.

WHAT IT ENCODES

Barium esophagram shows the classic 'bird's-beak' tapering with a dilated proximal esophagus; it is SUGGESTIVE but not diagnostic by itself. Manometry is still required to confirm. Barium also estimates emptying for follow-up.

BOARD TRAP

'barium bird's-beak alone confirms achalasia' -> wrong; it is suggestive only; manometry is the confirmatory gold standard.

7. Scope to exclude pseudoachalasia

WHAT YOU SEE

Gold-star banner 'SCOPE TO EXCLUDE' with 'PSEUDOACHALASIA' figure.

WHAT IT ENCODES

Always perform upper ENDOSCOPY to exclude pseudoachalasia: a GE-junction/cardia malignancy or infiltrating tumor mimicking achalasia (also Chagas disease in endemic areas). Suspect it with rapid symptoms, older age, marked weight loss.

8. Pseudoachalasia (malignancy/Chagas)

WHAT YOU SEE

Sinister hooded figure labeled 'PSEUDOACHALASIA' with a falling balance/scale.

WHAT IT ENCODES

Pseudoachalasia: secondary causes mimicking achalasia, chiefly GE-junction adenocarcinoma, and Chagas (Trypanosoma cruzi destroys myenteric plexus). Red flags: age >55, short symptom duration <6-12 months, profound weight loss. Endoscopy +/- imaging/EUS to exclude.

9. CCB / nitrates weak (RED X as best)

WHAT YOU SEE

Two pill/spray bottles 'CCB NITRATES' with a red X.

WHAT IT ENCODES

Calcium-channel blockers (nifedipine) and nitrates only transiently and weakly lower LES pressure; they are last-resort for patients who cannot have any procedure, NOT definitive therapy. Choosing them as best treatment is wrong.

BOARD TRAP

'CCB/nitrates as definitive achalasia treatment' -> wrong; they are weak/temporary, reserved for non-procedure candidates.

10. Botox ~6 months

WHAT YOU SEE

Box labeled 'BOTOX 6 MONTHS'.

WHAT IT ENCODES

Endoscopic botulinum toxin injection into the LES relaxes it but the effect lasts only ~6-12 months and wanes with repeats. Reserved for elderly or poor surgical/dilation candidates as a temporizing measure.

11. Definitive: dilation / Heller / POEM (gold star)

WHAT YOU SEE

Stacked blocks 'PNEUMATIC DILATION / HELLER MYOTOMY / POEM' with gold-star 'DEFINITIVE' banner.

WHAT IT ENCODES

Definitive therapies disrupt the LES: pneumatic balloon dilation, laparoscopic Heller myotomy (usually + partial fundoplication), or per-oral endoscopic myotomy (POEM). All carry esophageal perforation risk. POEM and Heller give the most durable relief.

12. Perforation risk

WHAT YOU SEE

Label 'PERFORATION RISK' with an endoscopic tunnel and scalpel.

WHAT IT ENCODES

The main complication of pneumatic dilation, Heller myotomy and POEM is esophageal perforation; POEM and post-myotomy patients also get post-procedure GERD. Watch for chest pain, fever, subcutaneous emphysema after the procedure.